

ST ANTHONY, NL, Canada

WMO: 718194

Lat: 51.370N

Lon: 55.600W

Elev: 12

StdP: 101.19

Time Zone: -3.50 (NAF)

Period: 86-95

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-26.1	-24.0	-31.5	0.2	-25.8	-29.2	0.3	-23.7	18.4	-7.1	16.8	-5.6	7.4	270	0.962

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.9	20.6	16.0	18.8	15.0	17.2	14.2	16.8	19.3	15.7	17.9	14.6	16.6	7.8	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.9	11.3	17.6	14.6	10.4	16.9	13.5	9.7	16.0	47.8	19.2	44.3	17.8	41.2	16.6	20.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.7	13.2	11.5	DB	-28.9	24.7	1.5	1.3	-29.9	25.7	-30.8	26.5	-31.6	27.3	-32.7	28.2	(4)
(5)			WB	-29.1	19.3	1.5	1.3	-30.2	20.2	-31.0	21.0	-31.9	21.7	-32.9	22.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.1	-12.4	-12.9	-7.6	-1.9	2.4	6.7	10.7	12.5	8.8	3.7	-2.1	-7.7		
	DBStd	9.48	6.14	6.16	5.38	3.32	3.39	3.59	3.66	3.14	2.88	3.04	4.08	5.46		
	HDD10.0	3798	694	641	545	357	237	112	36	11	58	196	363	548		
	HDD18.3	6659	953	874	803	607	493	350	237	182	287	454	613	807		
	CDD10.0	183	0	0	0	0	2	12	59	89	21	1	0	0		
	CDD18.3	2	0	0	0	0	0	1	1	1	0	0	0	0		
	CDH23.3	5	0	0	0	0	0	2	1	2	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	6.5	7.8	7.7	7.5	6.5	5.9	5.6	5.2	5.5	5.9	6.0	6.9	7.3		
	PrecAvg	1080	111	93	82	62	70	76	90	93	93	99	103	108		
(15) Precipitation	PrecMax	1304	232	147	145	107	144	153	177	192	197	185	198	161		
	PrecMin	894	46	48	34	20	17	30	36	29	32	41	42	51		
	PrecStd	100	41	28	28	21	31	26	30	32	30	35	34	32		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.9	2.0	3.8	7.6	16.4	20.7	22.7	23.5	18.4	13.0	8.1	2.5		
		MCWB	2.7	1.9	2.4	5.7	9.1	14.9	17.2	18.0	14.3	10.8	7.7	2.3		
	2%	DB	0.0	0.3	2.1	4.9	12.1	16.7	20.4	21.0	16.4	10.7	5.7	1.0		
		MCWB	-0.4	0.0	1.6	3.1	7.6	12.9	16.0	16.4	13.2	9.6	5.1	0.4		
	5%	DB	-1.5	-1.3	0.9	3.5	9.7	14.9	18.7	19.1	14.9	9.4	4.0	0.2		
		MCWB	-2.0	-1.7	0.4	1.8	6.5	11.6	14.9	15.6	12.1	8.3	3.3	-0.2		
	10%	DB	-3.0	-3.4	0.0	2.1	7.6	13.1	17.1	17.5	13.4	8.1	2.9	-0.6		
		MCWB	-3.5	-4.0	-0.5	0.9	5.4	10.4	14.0	14.7	11.2	6.9	2.3	-1.1		
	0.4%	WB	2.8	1.9	2.8	6.0	10.3	15.9	18.9	18.9	15.2	11.9	8.0	2.4		
		MCDB	2.9	1.9	3.2	7.7	13.9	19.6	20.3	21.9	17.3	12.4	8.1	2.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-0.3	0.1	1.7	3.6	8.4	13.9	16.9	17.2	13.8	10.0	5.4	0.6		
		MCDB	-0.2	0.3	2.1	4.4	11.2	15.8	19.4	19.5	15.7	10.5	5.5	0.8		
	5%	WB	-1.9	-1.8	0.5	2.0	6.9	12.4	15.6	16.0	12.6	8.5	3.5	-0.1		
		MCDB	-1.6	-1.4	0.9	3.1	9.2	14.2	18.0	18.0	14.3	9.1	3.9	0.1		
	10%	WB	-3.6	-3.9	-0.4	1.0	5.5	10.5	14.3	15.0	11.5	7.0	2.4	-1.0		
		MCDB	-3.1	-3.4	0.0	1.8	7.6	12.8	16.6	16.8	13.0	7.9	2.8	-0.7		
	5% DB	MDBR	7.9	7.3	6.8	4.8	6.1	7.6	7.1	6.9	6.5	5.1	5.1	6.5		
		MCDBR	9.9	7.9	6.6	6.2	10.1	11.4	10.5	9.0	8.9	6.8	5.6	6.2		
		MCWBR	10.1	7.9	6.6	4.9	6.7	8.0	6.4	5.4	6.2	5.9	5.7	6.5		
		MCWBR	10.1	7.6	6.6	5.5	9.3	11.0	9.3	8.2	8.1	6.3	5.6	6.5		
(35) Mean Daily Temperature Range	5% WB	MCWBR	10.4	7.8	6.7	4.6	6.9	8.5	6.7	6.3	6.3	6.3	6.0	6.8		
	Clear-Sky Solar Irradiance	taub	0.283	0.289	0.308	0.349	0.373	0.391	0.406	0.379	0.358	0.327	0.299	0.290		
		taud	2.320	2.283	2.270	2.260	2.349	2.364	2.372	2.443	2.478	2.521	2.478	2.316		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	735	837	889	888	881	865	844	851	831	794	725	657		
		Edh at Noon	63	86	106	121	117	117	114	101	87	69	55	55		
(44) All-Sky Solar Radiation	RadStd		0.80	1.74	3.01	4.20	4.79	5.04	5.00	4.21	3.12	1.85	0.96	0.61		
			0.10	0.24	0.43	0.48	0.56	0.44	0.43	0.27	0.25	0.17	0.09	0.07		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	+1.44	N/A	+0.37	+0.55	N/A	N/A	N/A	N/A

Nomenclature: See separate page